1 A student investigates the heating of a solid metal cylinder. Fig. 1.1 shows the cylinder of cross-sectional area A and height h.

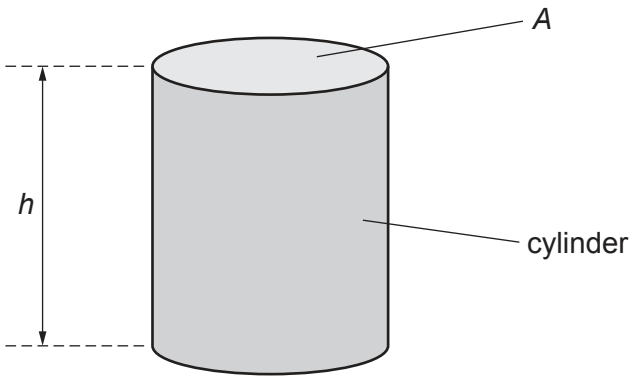

Fig. 1.1

The student places the cylinder and an electrical heater in a beaker of water. The electrical heater is switched on and the student measures the time t for the temperature of the water to increase by $\Delta\theta$.

A number of cylinders of the same material but with different cross-sectional areas are available.

It is suggested that the relationship between t and A is

$$Pt = AhW\Delta\theta + Z\Delta\theta$$

where P is the power of the heater and W and Z are constants.

Design a laboratory experiment to test the relationship between t and A.
Explain how your results could be used to determine values for W and Z.

You should draw a diagram, on page 3, showing the arrangement of your equipment.
In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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